Personal AI Collaboration During Ongoing Activities

Research interests and preliminary PhD directions

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Research motivation

I want to study personal AI collaborators that people can talk with while continuing an activity: assistants that relate conversation to what the person sees and does, reason with them about alternatives, carry decisions into useful digital actions, and become more personally useful through repeated collaboration. Their presence could be a voice, a glanceable annotation, a spatial model or a completed operation. My longer-term interest includes familiarity and shared experience. I would begin with a bounded activity and an implemented system, studying how these capabilities work together.

As people act, relevant objects, task state and intentions may change before they finish describing them. Previous decisions may help, yet their applicability can change. I am interested in interaction that lets a person and assistant maintain an evolving understanding, examine options together, and connect agreed next steps to outcomes they can inspect. Heads-Up Computing provides a direct setting because this collaboration must coexist with perception, movement and attention to the real world.

Connections to Synteraction Lab

TOM identifies user and environment understanding, multimodal interaction and activity-oriented services as foundations for wearable assistance, and demonstrates initial services [1]. It provides a possible implementation reference for connecting contextual input to assistance. An interaction contribution would still need to be established and evaluated.

CRAFT connects real-world encounters to ongoing fiction creation and maintains creative context across sessions [2]. PersonaMail supports intent articulation, granular revision and reuse of context-sensitive communication preferences [3]. These works motivate my interest in collaboration that develops through use, while showing that continuity and personalisation are existing foundations. PrISM also demonstrates how mixed-initiative dialogue can refine a procedural task assistant's understanding of the user's current step and context [4]. I would investigate a specific interaction difficulty within this established research space.

Candidate research questions

RQ1: How can voice, relevant egocentric context and interaction history support a shared and revisable understanding of a person's current activity?

RQ2: How can an assistant connect discussion of alternatives to useful digital actions while the person continues acting, and repair misunderstandings through subsequent interaction?

RQ3: Across repeated collaboration, which decisions and preferences should carry forward, and how should they be reconsidered when the current situation differs?

These questions describe an initial direction, not an established research gap. The first study would address one recurrent breakdown identified through observation. My architecture and CAVE experience prepares me to study people, environments and interactive systems, with observational work guiding the choice of activity.

Selecting an activity through observation

I would observe one recurring everyday activity and use replay interviews to understand how people explain intentions, refer to their surroundings, compare alternatives and recover from misunderstandings. Selection would require that people continue acting in the real world, environmental information matters to assistance, conversation connects to a small number of digital operations, and similar episodes recur. The activity should offer observable outcomes and feasible recruitment. This would establish where collaboration fails and whether a proposed interaction mechanism addresses a meaningful need.

A bounded situated prototype

The prototype would support the same selected activity with brief speech, relevant egocentric input and pointing where useful. It would connect these inputs to a limited set of digital tools appropriate to that activity, record their outcomes, and incorporate subsequent user corrections. For a proposed action, it would present relevant alternatives and the evidence supporting them, then use the observed outcome to inform the next exchange. I would explore a wearable configuration for receiving context while the person continues acting. Desktop replay would support debugging, but study episodes would retain environmental changes and the ongoing primary activity.

A candidate mechanism is a lightweight, inspectable account of the current goal, relevant activity state and pending action, updated through dialogue and observed outcomes. Voice and glanceable output could let the person correct an object reference, question a proposed alternative or revise a decision. Its form would follow the observed difficulty. I would test whether this shared account improves collaboration; additional memory or an extra display would not be presumed beneficial.

Comparison and evaluation

The principal baseline would be a capable conversational assistant with the same model, contextual inputs, history, digital tools and comparable processing budget. Both systems would support clarification, corrections and appropriate previews, confirmation and undo. The added shared account and revision workflow would initially be compared as a complete interaction design. A separate controlled comparison would be needed to attribute an effect to its representation alone.

Outcomes would include correctness of the assistant's activity interpretation, misunderstanding-repair effort, actions inconsistent with current requirements, repeated explanation burden, and time to useful progress. Task outcome quality would be assessed using criteria appropriate to the activity. Workload and interference with the ongoing activity would accompany perceived control and collaboration experience. Repeated sessions would examine appropriate reuse and mistaken carry-over of decisions or preferences. Sample size and duration would follow pilot variability and the primary outcome.

Preparation and scope

Game Fellows gave me experience with a rule-based Pygame collaborator, action-linked cues and controlled study preparation. My CAVE internship involved immersive-system integration; my capstone involved research design, reproducible evaluation and error analysis. Wearable integration and real-time multimodal interaction are skills I would develop. No participant study of this proposal has been conducted. The intended outcome is evidence about a bounded collaboration mechanism; long-term relationship formation remains a broader motivation. Ethics review would precede recruitment, with task-limited recording and disclosed external model processing.

References

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[2] Runze Cai et al. (2026). CRAFT: Exploring Wearable Creative AI on Smart Glasses for Fiction Writing in Real-World Contexts. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 10(3), Article 81. https://doi.org/10.1145/3831952

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